DANY HAIKAL BIN YAHYA

Bachelor of Computer and Communication System Engineering with Honours, UPM

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SUMMARY

Results-driven final-year Computer & Communication Systems Engineering (Hons.) student at Universiti Putra Malaysia with hands-on experience across AI/ML, IoT & Embedded Systems, and Networking. Built end-to-end systems using Jetson Nano, ESP32, YOLOv12, and LSTM from environmental forecasting to RFID inventory tracking. Available for any graduate trainee or full-time engineering roles where AI, IoT, firmware, hardware, software and leadership intercept.

EDUCATION

Universiti Putra Malaysia - Bachelor of Computer and Communication Systems Engineering with Honours — Selangor, Malaysia
CGPA: 3.43 / 4.000 - Second Upper Class Honour — **Oct 2022 - Sep 2026**

Centre of Foundation Studies Universiti Putra Malaysia – Foundation in Agricultural Science — Selangor, Malaysia
CGPA: 3.55/4.00 – First Class Honour — **Aug 2021 – Jun 2022**

PROJECT INVOLVEMENT

Intelligent Poultry & Feed monitoring System to Detect Feed Level and Predict Optimize Feed Volume

- Designed the data collection setup before development started using Jetson Nano, Pi Camera, and Iot Sensor to read the temperature and humidity data.
- Collected and annotated data before feeding it to AI for training, validation and testing.
- Trained and fine tuned the AI model hyperparameter and tested it using sample data to optimize AI accuracy without overfitting.
- Validated and evaluated the AI model performances

IoT - CO2 and MoldRisk Prediction System Using AI

- Deployed and configured an IoT monitoring system using environmental sensors to collect environmental data
- Developed IoT data pipelines to process, store, and manage real-time sensor data efficiently.
- Processed sensor data using Python and KNN-based data imputation techniques, and trained an LSTM AI model for 30-minute prediction of MoldRisk and CO2 levels.
- Integrated Flask API with the system to automate prediction data transfer and visualize real-time readings and AI predictions on an interactive dashboard.

RFID Smart Inventory Management System

- Led the system design process, focusing on hardware selection, data flow architecture, and user interface optimization to ensure efficient device functionality and usability.
- Designed and developed a warehouse RFID detection system using ESP32, PN532 RFID reader, and LCD touchscreen display for real-time inventory tracking.
- Developed and simulated the system backend and frontend to manage data operations including sorting, adding, deleting, and processing scanned records.

Stray Dog Detection System using Artificial Intelligence

- Led project planning and task distribution to ensure an efficient and organized AI system development process.
- Collected, preprocessed, and managed datasets from multiple platforms for AI model training and validation.
- Trained, and evaluated a YOLOv12 AI model for real-time stray dog detection based on project objectives and research analysis.
- Developed a Streamlit dashboard integrated with map visualization and alert notifications for real-time stray dog detection monitoring.

EXPERIENCES

Beyond2U Sdn. Bhd. **Selangor, Malaysia**
System Engineer Intern Jul 2025 - Oct 2025

- Execute tech refresh replacement following provided guidelines on several AmBank branches, while troubleshooting any expected or unexpected problems.
- Setup new ip and domain configurations to ensure the network is private, so that no external party can access the data inside the AmBank domain.
- Transferred all the required data from the old pc to the new one without missing any single file.
- Installed BitLocker on the PC so no unrecognized device can connect to the PC through wired connection.
- Executed KillDisk on the old PC to hard disk to make sure no possible way the data can be leaked.
- Deal with AmBank Branch Manager and user for arrangement on PC deployment to avoid unproductive and inefficient work flow.

TECHNICAL SKILLS

- **Programming Languages :** Python, Java, C++, HTML, Verilog
- **Engineering & Design Software :** MATLAB, Simulink, AutoCAD, SolidWorks, Microsoft Visio, Microsoft Project
- **Networking & Communication Tools :** GNS3, OptiSystem, MQTT, Oracle VirtualBox
- **FPGA & Embedded Systems :** Intel Quartus Prime
- **Internet of Things (IoT) & Data Visualization :** Node-RED, Grafana, Streamlit
- **Database Management :** MySQL, InfluxDB
- **Artificial Intelligence & Machine Learning :** YOLO, LSTM, Neural Networks, Google Colab, Ollama
- **Productivity Tools :** Microsoft Excel, Microsoft Word, Microsoft PowerPoint

LEADERSHIP EXPERIENCES

Universiti Putra Malaysia **Selangor, Malaysia**
Participants in Build Your Own Machine Learning Workshop 2024

- Participated in a hand-on “Build Your Own Machine Learning” workshop, gaining practical experience in building, training, and evaluating machine learning models

Universiti Putra Malaysia **Selangor, Malaysia**
Wire Bond Course Introduction & Career Talk, organized by NXP Malaysia 2024

- Participated in a specialized course on wire bonding technology, learning about key processes, materials, and quality considerations in semiconductor manufacturing

Taman Negara Malaysia **Pahang, Malaysia**
CSR Project: Unity Through STEM 2024

- Led my team in introducing ferrofluid to the native people and give them exposure about STEM

Universiti Putra Malaysia **Selangor, Malaysia**
CSR Project: Rimba ke Menara Gading 2023

- Led Bateq native people in exploring urban side of Malaysia at Kuala Lumpur while ensuring their safety

REFERENCES

Prof. Dr. Mohd Adzir bin Mahdi
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Universiti Putra Malaysia
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